NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

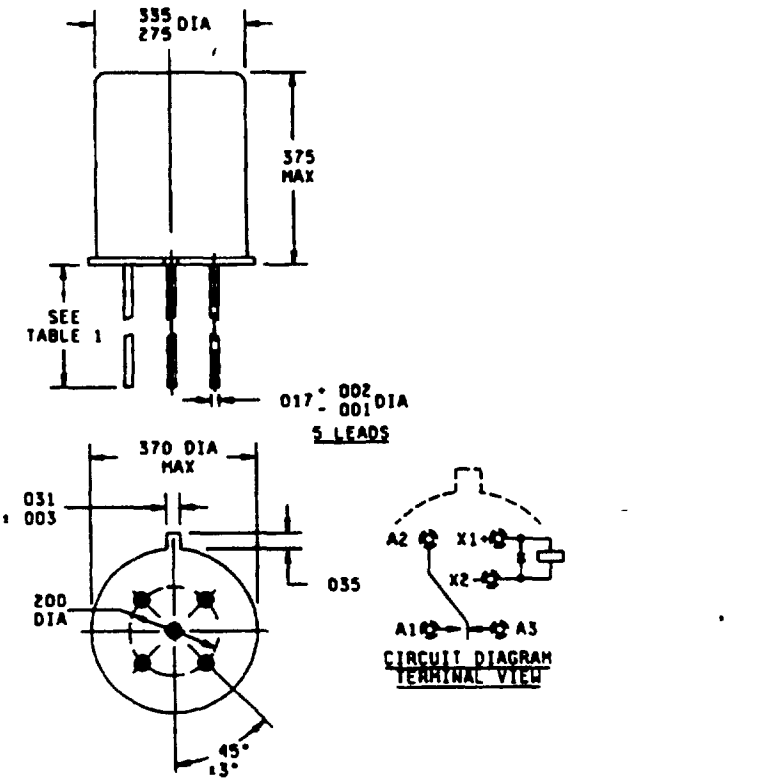
MIL PRF 19016/25F
20 JULY 1988
SUPERSEDING
MIL-R-39016/25D
10 February 1984

PERFORMANCE SPECIFICATION SHEET

RELAYS ELECTROMAGNETIC ESTABLISHED RELIABILITY, SPDT, LOW LEVEL TO 1 0 AMPERE (SENSITIVE, 40 MILLIWATTS) WITH AN INTERNAL DIODE FOR COIL TRANSIENT SUPPRESSION

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of MIL-R-39016



Inches	mm	Inches	mm	Inches	mm
001	0 03	031	0 79	335	8 51
002	0 05	035	0 89	370	9 40
003	0 08	200	5 09	375	9 53
017	0 43	275	6 99		

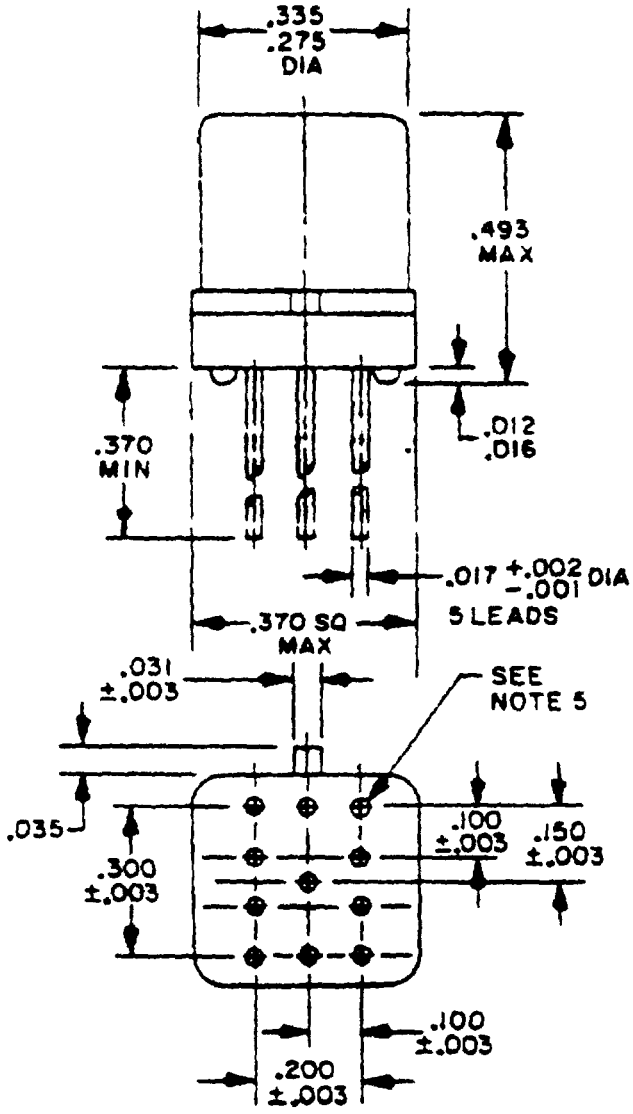
NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified tolerance is ± 010 (0 25 mm)
4 Terminal numbers shown above are for reference only Numbers do not appear on the relay
5 Relays shall have a plus (+) sign placed on the circuit diagram as shown
6 (-) symbol optional in accordance with MIL-STD-1285
7 Circuit diagram shown as part of the terminal view

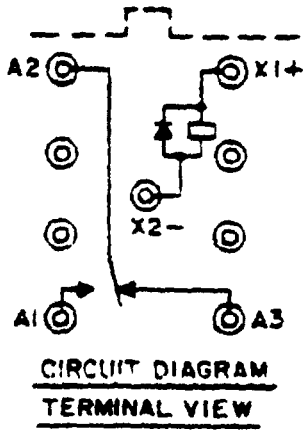
FIGURE 1 Dimensions and configuration

(E) denotes changes

DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

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CIRCUIT DIAGRAM
TERMINAL VIEW

Inches	mm
.001	0.03
.002	0.05
.003	0.08
.012	0.30
.016	0.41
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.150	3.81
.200	5.08
.275	6.99
.300	7.62
.335	8.51
.370	9.40
.493	12.52

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
(E) 4. Spreader pads shall be certified to MIL-M-38527, M38527/05-004.
5. Dimensions and tolerance shown for the bottom view of the spreader pad are for the center to center locations of the holes in the spreader pad.
6. Shape optional within the envelope dimension.
7. Terminal numbers shown above for reference only. Numbers do not appear on relay.
8. Relays shall have a (*) sign placed on circuit diagram as shown.
9. All leads shall be electrically insulated from the case
coil symbol optional in accordance with MIL-STD-1285
11. Circuit diagram shown on part is the terminal view.

FIGURE 1. Dimensions and configuration relay with spreader pad attached

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REQUIREMENTS:

CONTACT DATA

Load ratings

High level (relay case grounded):

Resistive.

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 Hz case not grounded.

250 milliamperes at 115 V ac 60 Hz case not grounded.

100 milliamperes at 115 V ac 60 and 400 Hz case grounded.

Inductive load. 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp 0.10 ampere at 28 V dc.

Low level: 10 to 50 uA at 10 to 50 mV dc or peak ac.

Intermediate current. Applicable.

Contact resistance or voltage drop.

(E) Initial: 0.10 ohm maximum (0.125 ohm maximum with spreader pad attached).

High level:

During life: Not more than 5 percent of open circuit voltage.

After life. 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Low level:

During life. 33 ohms maximum.

After life: 0.15 ohm maximum (0.175 ohm maximum with spreader pad attached).

Intermediate current:

During. 1 ohm maximum.

After. 0.20 ohm maximum (0.225 ohm maximum with spreader pad attached).

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").

Contact stabilization time. 2.0 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only) Two times rated current

COIL DATA: See table I

Operate time 4.0 ms maximum over temperature range with rated coil voltage

Release time 7.5 ms maximum over temperature range from rated coil voltage.

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ELECTRICAL DATA

Insulation resistance 1/ 10,000 megohms minimum at 500 V dc, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum

Dielectric withstanding voltage 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	300 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	---	
Between coils of dual coil relays	---	

(E) DIODE CHARACTERISTICS. 2/

Maximum transient voltage 1 volt

(E) Coil transient suppression: Applicable.

(E) Semiconductor in-process screening: Applicable, visual inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074

ENVIRONMENTAL DATA

Temperature range: -65°C to +125°C.

(E) Vibration (sinusoidal): MIL-STD-202, method 204 Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

(E) Vibration (random): MIL-STD-202, method 214, test condition IG Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts (applicable to qualification and group C testing only).

(E) Shock (specified pulse) MIL-STD-202, method 213, test condition B (75 g's) Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable

Acceleration: Applicable.

Salt atmosphere (corrosion): In accordance with MIL-STD-750, method 1041.

1/ Insulation resistance and dielectric withstanding voltage tests must always precede all other specified electrical measurements. Connect all coil terminals together to avoid damage to diodes

(E) 2/ WARNING Reverse polarity on coil terminals will destroy diode

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PHYSICAL DATA

Terminal strength MIL-STD-202, method 211

Pull test: Test condition A, 1 pound pull.

Bend test Test condition C, 1/2 pound load.

(E) Twist test: As specified in MIL-R-39016.

Solderability Applicable.

Dimensions and configuration See figures 1 and 2.

Weight: 2.84 grams (0.10 ounce) maximum, 3.09 grams (0.109 ounce) maximum with spreader pad attached.

Seal Hermetic.

(E) Minimum marking. Military part number, "J" with the date code (example J8530), circuit diagram, and manufacturer's name or source code.

LIFE TEST REQUIREMENTS

High level: 100,000 cycles per relay.

(E) Low level: 100,000 cycles plus 900,000 cycles mechanical life

PART NUMBER: M39016/25- (dash number from table I and suffix letter designating failure rate level).

(E) TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/				Coil voltage (V dc) 3/		At 25°C				Over temperature range		
Lead length 1.500 min 4/	Lead length .187 ±.010	Lead length .500 min	Relays with spreader pads (fig. 2) 5/	Rated	Max	Coil resistance ohms ±10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified drop-out value (voltage) (V dc)
017	025	033	041	5.0	8.0	125	2.8	1.7	0.23	3.7	2.4	0.15
018	026	034	042	6.0	11	255	3.5	2.0	0.28	4.5	2.8	0.18
019	027	035	043	12	22	1,025	7.0	4.0	0.63	9.0	5.6	0.40
020	028	036	044	26.5	45	4,000	14.2	8.0	1.37	18	10.4	0.89
021	029	037	045	32	57	6,500	18.7	10.6	1.59	24	15	1.0
022	030	038	046	40	75	11,000	23.3	13.3	2.0	30	18.7	1.3
023	031	039	047	9.0	16	630	5.3	3.0	0.54	6.8	4.2	0.35
024	032	040	048	18	33	2,300	10.5	6.0	0.91	13.5	8.4	0.58

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles). L, 3.0, M, 1.0, P, 0.1, R, 0.01. Example, 017L - - - - - 048R.

3 CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

4 1.500 leads are inactive for new design

5 Relays supplied with spreader pads (-041 through -048) shall have the pad rigidly attached

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QUALIFICATION INSPECTION

Qualification inspection and sample size: See table II.

TABLE II. Qualification inspection and sample size. 1/

Single submission		Group submission
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable	M39016/25-036	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable
	M39016/25-033 M39016/25-034 M39016/25-035 M39016/25-037 M39016/25-038 M39016/25-039 M39016/25-040	2 units each part number Qualification inspection, group II

1/ For retention of qualification or extension of qualification to lower failure rate levels, all life test data accumulated on MIL-R-39016/21 and MIL-R-39016/26 may be used in addition to MIL-R-39016/25 data. Prior to performance of retention of qualification inspection, the relay manufacturer shall preselect the sampling plan

2/ The number of units required for qualification testing shall be increased as required in group V, table II, MIL-R-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification inspection, the relay manufacturer shall preselect the sampling plan.

Initial qualification of relays supplied with spreader pads (-041 through -048), shall be tested as specified below.

Perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below:

Before installation of pad, screening, visual and mechanical examination (internal), thermal shock, resistance to solvents, vibration (sinusoidal), vibration (random), shock (specified pulse), acceleration, terminal strength, magnetic interference (when specified), capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016, in the order shown below.

Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Qualification inspection (reduced testing for previously qualified relays) for relays supplied with spreader pads (-041 through -048), two (2) units of the 26.5 volt rated coil voltage (-044) shall be tested as specified below:

Before installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below.

For failure rate level L only Screening
For failure rate levels M, P, and R Vibration (sinusoidal) test duration shall be 10 minutes, vibration (random), particle impact noise detection (PIND, when specified), screening

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After installation of pad perform the following tests as specified in the qualification inspection table of MIL-R-39016 in the order shown below:

(E) Insulation resistance, dielectric withstanding voltage, static contact resistance, specified pickup, hold, and dropout values (voltages), coil resistance, operate and release time, contact dynamic characteristics, coil transient suppression (when specified), solderability, seal, visual and mechanical inspection (external).

Group A testing for relays supplied with spreader pads (-041 through -048) shall be tested as specified below:

(E) Before installation of pad perform subgroup 2 of group A tests.
After installation of pad perform subgroups 3 and 4 of group A tests.

Qualification inspection (reduced testing) and sample size. See table III
If the relays produced for MIL-R-39016/25 are similar in construction and design except for the diodes and headers, as applicable, to the relays produced for MIL-R-39016/21 or MIL-R-39016/26, then reduced testing for qualification of MIL-R-39016/25 relays may be performed concurrent with or subsequent to successful qualification of MIL-R-39016/21 or MIL-R-39016/26.

(E) TABLE III. Qualification inspection (reduced testing)

Examination or test
2 units each coil voltage Group II of qualification inspection table 1 unsealed sample unit for internal examination

SUPERSESSION DATA:

Supersession data: See table IV.

TABLE IV. Supersession data. 1/

Superseded part no M39016/25-	New part no. M39016/25- 1/	Superseded part no M39016/25-	New part no. M39016/25-
001	017	009	027
002	018	010	028
003	019	011	029
004	020	012	030
005	021	013	023
006	022	014	024
007	025	015	031
008	026	016	032

1/ Dash numbers -017 through -024 are inactive for new design and are for support of existing equipment designs only.

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Cross reference for Government logistical support: See table V.

TABLE V Cross reference for Government logistical support.

Superseded part number M39016/25-	New part number M39016/25-	Support with part number M39016/	New part number M39016/25-	Support with part number M39016/
001	017	25-017	033	25-033
002	018	25-018	034	25-034
003	019	26-019	035	26-035
004	020	26-020	036	26-036
005	021	26-021	037	26-037
006	022	26-022	038	26-038
007	025	25-033	039	26-039
008	026	25-034	040	26-040
009	027	26-035	041	25-041
010	028	26-036	042	25-042
011	029	26-037	043	26-043
012	030	26-038	044	26-044
013	023	26-023	045	26-045
014	024	26-024	046	26-046
015	031	26-039	047	26-047
016	032	26-040	048	26-048

CONCLUDING MATERIAL

Custodians.
Army - ER
Navy - EC
(2) Air Force - 85

Review activities
Army - AR
Navy - AS, OS, SH
(3) Air Force - 99
DLA - ES

User activities:
Navy - MC
(2) Air Force - 11

Preparing activity
Navy - EC

Agent:
DLA - ES

(Project 5945-0757-19)

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